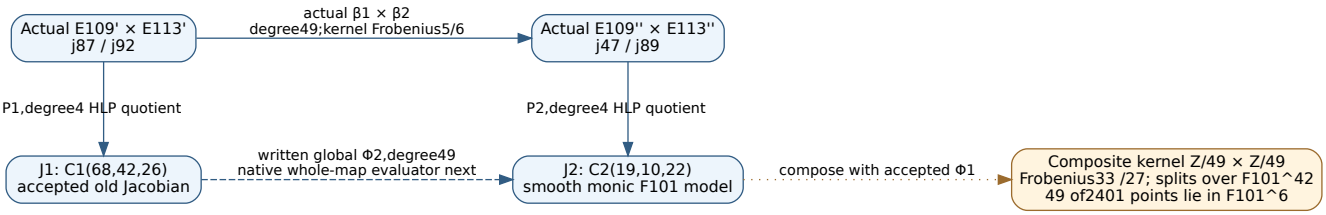
The next nondual seven-transfer

2026-11-11 · CT24 fixed F101 preflight and written global construction

The next actual elliptic degree7 maps preserve the Frobenius5/6 directions and give new targets (59,27),j47 and (23,7),j89. Their transported2-gluing produces the smooth monic C2:$y^2=x^6+22x^4+10x^2+19$. A written HLP quotient argument constructs the global degree49 map $\Phi 2: J1 \to J2$. The native full Jacobian evaluator and source/global runtime certificates are next. PCCb07c76/190ae8/e98755; scientific standing unsigned. [pcc:unverified]



Exact kernels, normalization and controls

The E109′=(26,58) kernel is [99,94,75,1], giving (59,27),j47. The E113′=(69,50) kernel is [53,19,40,1], giving (23,7),j89. Both previous-dual kernels are identified by exact polynomial equality and actual Frobenius2/4. The unique other directions have5/6; selection uses the kernel action rather than a fitted j-invariant. [pcc:unverified]

The reused exact constructor checks the curve identity and both global rational-x dual[7] identities. Every109/113 source and target point passes bijection, positive dual7, scalar and two-elliptic-hop reverse49 controls. The inverse49 scalars are89 and30. No new field/poly/Kohel engine is written. [pcc:unverified]

The raw HLP sextic is [45,0,29,0,84,0,13], with ordinate scale35 and normalized constant19, square root25. All coefficients descend to F101; smoothness, product L-polynomial and Jacobian order12317 pass. Actual quotient charts (destination,short shift,scale) are [(1,41,18),(0,37,18)], including the component swap. All120 curve points, x0/infinities, complete109/113 coordinate bijections and all three actual2-support matches pass. [pcc:unverified]

Global map and composite-kernel field boundary

Quotient construction and Jacobian distinction

For Ai=E109,i × E113,i, the HLP quotient Pi:Ai→Ji has degree4, Galois graph Γi and polarization norm2. Actual transport gives Γ2=β(Γ1), hence P2β kills Γ1 as a finite étale group scheme. The quotient universal property defines the regular F101 map Φ2P1=P2β, of degree 4·49/4=49. Pulling back polarizations gives Φ2Hat·Φ2=Φ2·Φ2Hat=[7]. Read SUCCESSOR_GLOBAL_PROOF.md/PCCe98755. [pcc:unverified]

HLP embeds both elliptic factors. The trace −7 connected elliptic subvariety is unique over F101 because the trace −7 and trace −11 factors are not isogenous over F101. The j-invariants87≠47 and12≠47 therefore distinguish J2 from J1 and J0 as unpolarized F101 abelian varieties. The rational109/113 components are preserved by degree coprimality. Executed all-class evaluation is still a native obligation. [pcc:unverified]

Why the full composite kernel needs a larger field

For each nondual β·α, image α(E[7]) is the old dual kernel. The new β-kernel meets it trivially, so Ker(βα)[7] has order7 within a degree49 kernel: this kernel is cyclic49. Frobenius reduces to5/6 and satisfies the source characteristic polynomial modulo49. Its unique lifts are33/27, with exact orders42/14 and sixth powers8/29. [pcc:unverified]

Thus Ker(Φ2Φ1) is Z/49 × Z/49, of order2401 and polarization norm49. All points split over $F101^{42}$, while only49 lie over $F101^{6}$. The maps retain prime-field coefficients. This is a concrete boundary for a complete geometric point list in the current degree6 representation; no degree42 engine or such point enumeration was introduced. [pcc:unverified]

Six CyclicFortyNine.lean finite-ring certificates pass the Lean kernel gate, with zero trusted native. They use decide, no native_decide or new axioms. PCC82b8fb; Tate/Cayley-Hamilton, elliptic cyclicity, HLP, polarization and subvariety uniqueness remain written bridges. Complete source census109 includes the earlier transfer helpers. [pcc:unverified]

Durable evidence and next work

Read CT24_PREFLIGHT.md, CT24_CHAIN_KERNEL_PLAN.md and the standalone successor-preflight crate. All five final tests08 and strictClippy07 pass; Clippy06 paired-loop failure/source05 is preserved and zip iteration fixes it without changed values. Independent flat-field replay also checks the previous Psi kernel TSV, with its own limited scope. [pcc:unverified]

Primary sources: HLP math/9809210v1 §3.2 Propositions3–4 and MilneAV Proposition10.20/II Theorem1.1. KB/atlas access is unavailable. Next implement Φ2/Phi2Hat, both complete domains and109-source/direct controls, then transport the actual R1 kernel and construct R2/Psi2. Ledger/admission/ budget/vault limits persist; no timing or computational gain is claimed. [pcc:unverified]